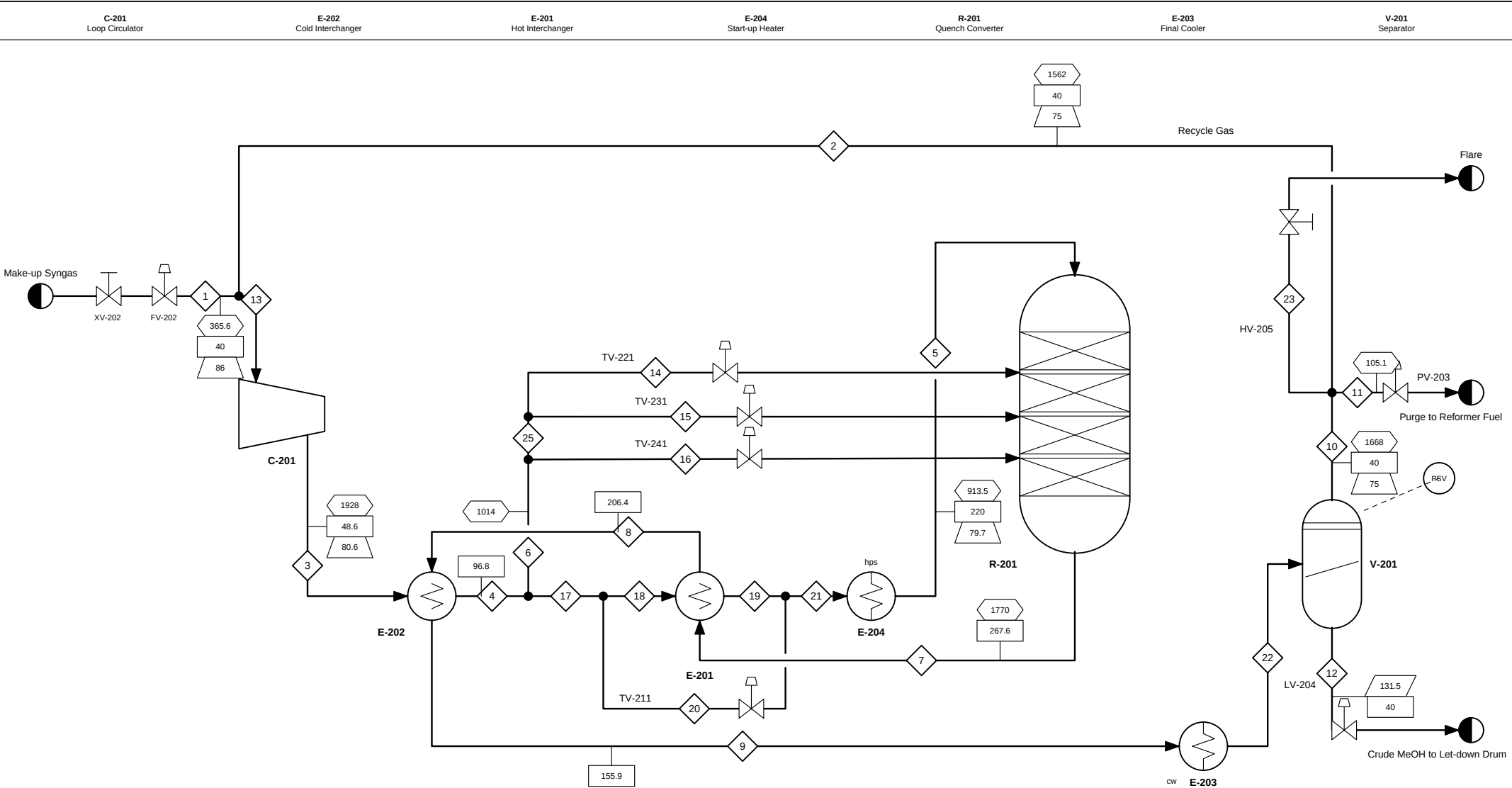




**REACTIONS (Cu/ZnO/Al<sub>2</sub>O<sub>3</sub>, 4 quench beds)**

CO<sub>2</sub> + 3 H<sub>2</sub> <=> CH<sub>3</sub>OH + H<sub>2</sub>O

CO<sub>2</sub> + H<sub>2</sub> <=> CO + H<sub>2</sub>O

Methanol 112.5 t/h; loop 75 bar g

**NOTES**

1. Gas flows in kNm<sup>3</sup>/h; liquid in t/h; T in °C; P in bar g. Design point of the simulator.

2. Stream numbers 1-12 as in the Process tab stream table; others carry no data.

3. Let-down drum, reformer fuel and flare are outside the battery limit.

4. PSV-V201 (set 88 bar g) protects the loop and relieves to flare; HV-205 is the manual blowdown.

|                                                                                                |                                                                                                                   |                                                                                                            |                                                                                                      |                                                                                                      |                                                                                                    |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>LEGEND</b>                                                                                  |  Gas Flow (kNm <sup>3</sup> /h) |  Mass Flow (t/h)        |  Temperature (°C) |  Pressure (bar g) |  Process Line |
|  Block Valve |  Control Valve                 |  Process Input / Output |  Stream Number    |  Line Crossing  |                                                                                                    |
|  Compressor  |  Heat Exchanger                |  Packed-Bed Reactor     |  Separator        |                                                                                                      |                                                                                                    |

|                                                   |           |              |                    |
|---------------------------------------------------|-----------|--------------|--------------------|
| OPERATOR TRAINING SIMULATOR - HAZID STUDY DRAWING |           |              |                    |
| <b>PROCESS FLOW DIAGRAM</b>                       |           |              |                    |
| METHANOL CONVERTER VER 1 - R-201 SYNTHESIS LOOP   |           |              |                    |
| DRAWING No.<br>PFD-201                            | REV.<br>0 | SIZE<br>A4   | SHEET<br>1 OF 1    |
| DRAWN<br>Apinan Sootitiantawat                    | CHECKED   | SCALE<br>NTS | DATE<br>2026-09-29 |